

Good/Bad Layout Board

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Abstract—The good/bad layout board serves as a demonstration of good and bad design practices and their influence on the performance of the board. Using the 555 timer built in the 555-timer lab, we create a clock signal to different hex-inverter circuits and measure the noise detected on major power lines and draw conclusions on what exactly the "best design practices" we learned in previous labs do for our boards. After completion of the lab, we find that the placement of decoupling capacitor influences the amount of noise on the power line whereas the presence of a return plane has more of an influence on the noise on the ground line.

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1. Introduction

When building a board, understanding why we make certain design choices and the consequences of making bad choices is imperative for optimized board design. This lab aims to put our understanding of good PCB design practices into a physical board by creating three designs of the hex-inverter clock circuit with varying degrees of quality.

2. Objective

- Construct a hand-drawn version of the board to gain a basic understanding of the functionality of the hex inverter.
- Create a solderless breadboard version of the board as a proof of concept
- Recreate the SBB three times using different design practices:
 - A circuit with no ground return plane with a badly placed decoupling capacitor (bad design)
 - A circuit with a ground return plane but a badly placed decoupling capacitor (less bad design)
 - A circuit with a ground return plane and a well-placed decoupling capacitor (good design)
- Perform CDR and order the board.
- Assemble the board and test for functionality.
- Take measurements from a given Good/Bad Layout board to use as a ground truth.

- Take measurements from our board and draw conclusions based on the design practices used and comparisons between our board and the ground truth.

3. Plan of Record

What does it mean to work?

- The board must be powered by the 5v power supply.
- All components must be powered by 3.3v from the LDO controlled by a switch.
- The timer must output a square waveform with a frequency of 500 Hz and duty cycle of 65.6% and power the output LED.
- All of the LEDs on the board must be powered regardless of design practices.
- The inverter circuits must output a square signal with a frequency of 400-500 Hz and a duty cycle of 40%.
- The quiet low and high signals should contain a significant amount of noise on both of the bad versions of the circuit and should not have any noise on the good version of the circuit.

Features of the Board:

- 2-layer board consisting of a signal layer and a common ground return plane
- 5v DC power jack
- 1206 components
- 3.9in. x 3.9in. board size

Risk Management:

- All components, test points, switches and sections are labeled for easy debugging and assembly.
- Power lines and signal have different trace widths (15mil and 6 mil respectively)
- Inverter circuits are made with the exact same topology (unless the design practice of interest demands otherwise) to ensure that differences are strictly due to design practices.
- 1206 components were used for easy soldering.
- 3.9in x 3.9in board size to minimize production cost

Components:

- 5v power jack x1
- 3.3v LDO x1
- Switch x4
- 22 μ F capacitors x2
- 1 μ F capacitor x1
- 22 pF capacitor x3
- 10k Ω resistors x3
- 1k Ω resistors x5
- 47 Ω resistors x9
- Red LED x12
- LMC555 Timer x1
- Hex Inverter (SN74HC14DR) x3
- Test Point x12

4. Concept

This circuit will take advantage of the fast switching signals of the 555 timer and the hex inverter to demonstrate the effect on the power line of the die in the hex inverter.

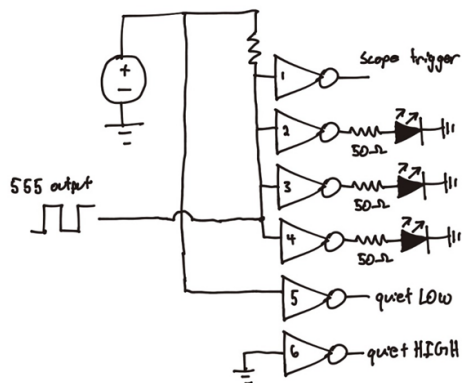
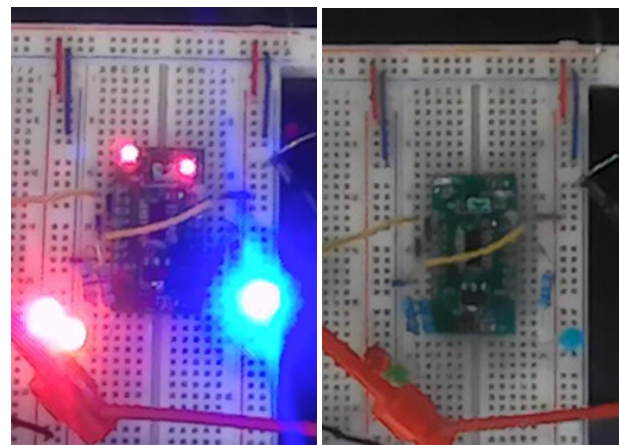


Figure 1. Napkin sketch of hex inverter circuit

Fig. 1 demonstrates the wiring of the hex inverter. A 555 clock signal will be sent into the input of NOT gates 1 to 4. Gates 5 and 6 will be used to measure the ground (quiet LOW) and VCC (quiet HIGH) and the die of the inverter. With this configuration, we should be able to monitor the output of the inverter and the voltage readings on the LOW and HIGH pins with the test points. With the LEDs providing a load, we should expect readable noise on the HIGH and LOW pins.

5. Solderess Breadboard Circuit



(a) SBB (Powered On) (b) SBB (Powered Off)

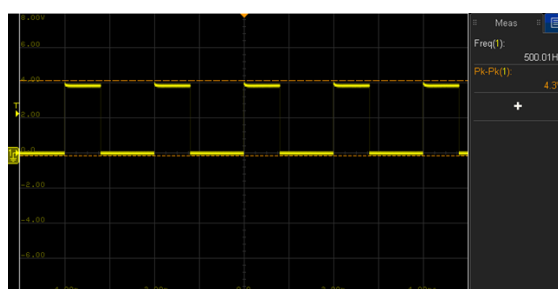
Figure 3. SBB version of fig. 1 napkin sketch

The circuit was then constructed on an SBB as a proof of concept. For the SBB circuit to work, we need to ensure the inverter got a 500 Hz signal from the 555 timer causing the LEDs to light up. Since we were able to generate such a signal from the 555-Timer report, we will reuse that circuit. We should measure a similar signal from the hex inverter, but with a 40% duty cycle. There should also be a readable voltage drop on the quiet LOW and HIGH on every switch of the signal.

Note: A function generator was used to generate a pulse signal. A frequency and duty cycle similar to that of the board from the 555-timer report was used to maintain consistency.

Following the schematic from fig. 1, The LEDs on inverters 2, 3, and 4 light up as expected the schematic in. After building the circuit and using a function generator to create a 500 Hz 60% duty cycle pulse wave, the LEDs light up as expected (fig. 3) with red, green, and blue indicating output from inverters 2, 3 and 4 respectively.

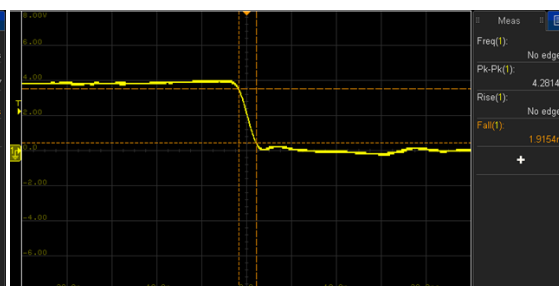
Analyzing the readings from the scope trigger and quiet LOW and



(a) Hex signal full waveform



(b) Hex signal rise time

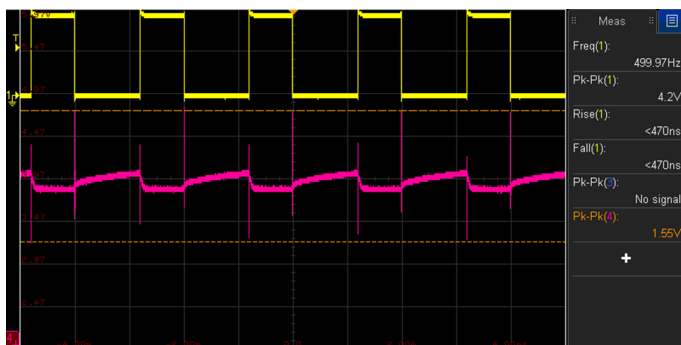


(c) Hex signal fall time

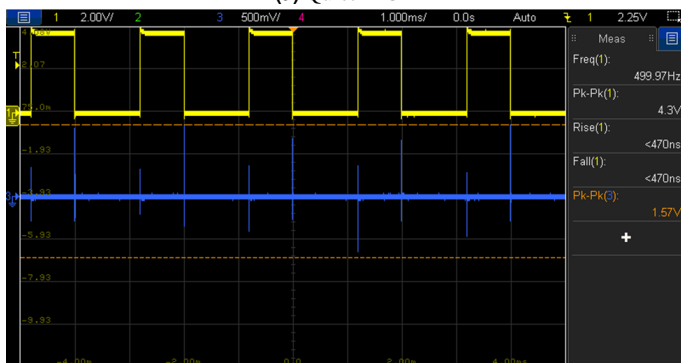
Figure 2. Hex inverter scope readings

HIGH (fig. 2), we get the expected readings with a fast enough rise and fall time ($>10\text{ns}$) to induce noise on the power lines.

Moving forward, the scope will be triggered on this waveform to better analyze the noise on the HIGH and LOW pins. Additionally, this waveform will be held at an offset so that it is easier to see what part of waveform induces noise.



(a) Quiet HIGH



(b) Quiet LOW

Figure 4. Readings on Quiet HIGH and LOW of hex-inverter chip

On the HIGH and LOW pins, we see a total peak-to-peak voltage drop of around 1.55 V for both HIGH and LOW lines. After proving the functionality of the SBB circuit and showcasing the noise generated on the HIGH and LOW, we can proceed to create the PCB while testing the effects of good PCB design practices on the LOW and HIGH signals.

6. Designing the PCB

While making the inverter circuit in Altium, three different designs were created for testing (fig. 5). One design with no return plane nor a well-placed decoupling capacitor, one with a return plane but without a well-placed decoupling capacitor, and one with both a return plane and well-placed decoupling capacitor.

For the 555-timer, the same topology used in the report was followed in this design since that design successfully created the pulse wave required for this board. Test points, LEDs and labels were included for all significant power and signal lines to test for functionality and allow for easy troubleshooting.

Additionally, to ensure that all other variables were as controlled as possible, the topology for the bad design and (less) bad design sections were kept exactly the same. This way, we can derive more robust conclusions regarding the effect of the design practice of interest.

After receiving, assembling, and powering the board, preliminary measurements are taken to ensure the board is properly receiving power.

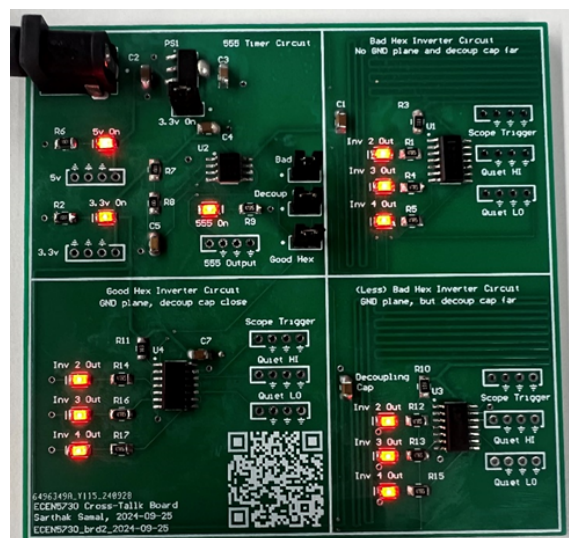
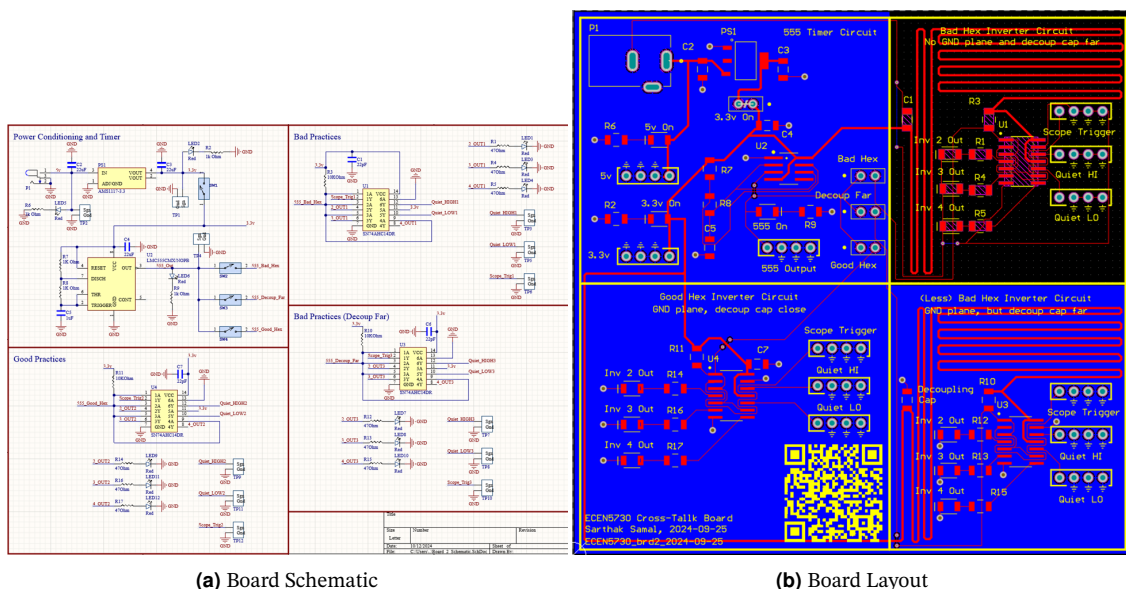


Figure 6. Assembled board, powered on

After power, all of the LEDs light up the same way (fig. 6) indicating that the board is properly receiving 5v from the plug and that all of



(a) Board Schematic

(b) Board Layout

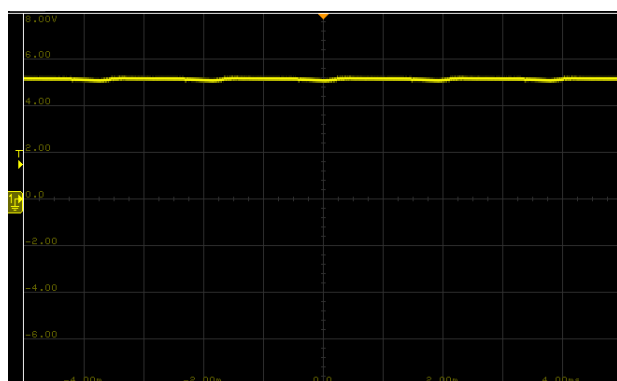
Figure 5. PCB Design in Altium Designer

the components are receiving 3.3v from the LDO. After verifying that the board gets power, we can move on to key measurements on the power rail.

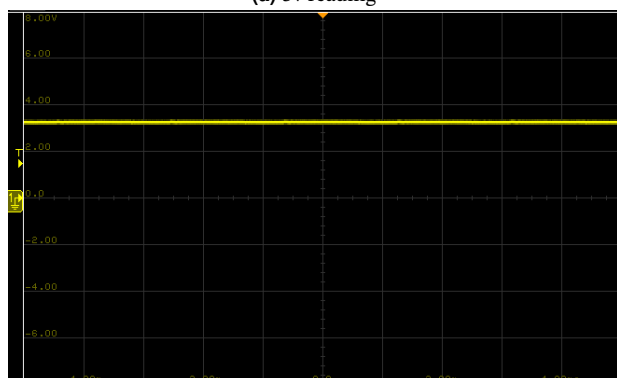
7. Key Measurements

7.1. Power Lines

When measuring the power lines of the board, we should expect to see little to no voltage drops since we have used decoupling capacitors to minimize noise.



(a) 5v reading



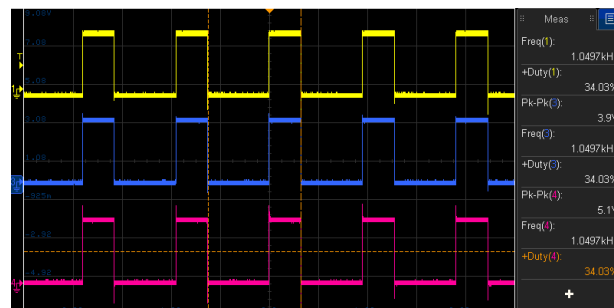
(b) 3.3v reading

Figure 7. Board power line readings (2V/div)

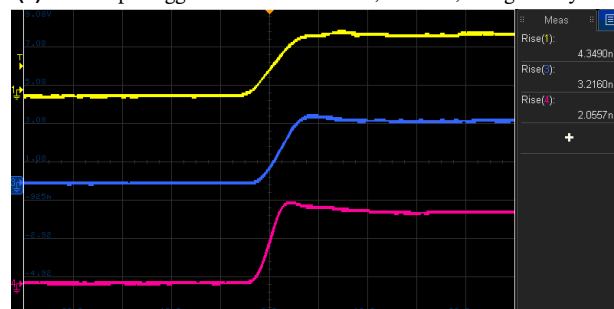
According to fig. 7 There is some noticeable fluctuation in the 5v line. These readings were taken with the entire board powered, meaning all of the LEDs were drawing power from the board's power line which is a likely cause for these fluctuations. However, since the 3.3v line powers all of the components on the board, a stable reading on the 3.3v line is more important for determining the functionality of the board. Since there is little to no fluctuation on the 3.3v, we can consider the board to be adequately powered.

Before taking measurements on our board, we first take measurements from a given Good/Bad Layout board to get an idea of what we should expect on our board.

7.2. Ground Truth Scope Trigger



(a) Full scope trigger waveforms for bad, less bad, and good layouts



(b) Rise time of scope trigger waveforms for bad, less bad, and good layouts

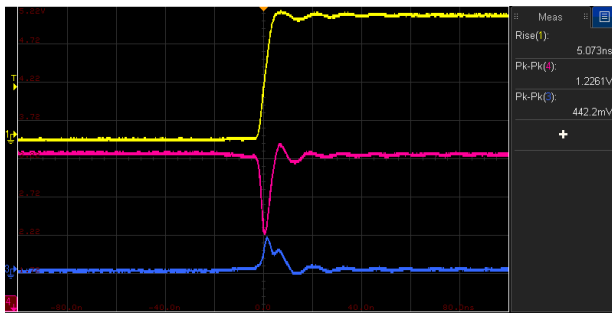
Figure 8. Scope trigger waveform characteristics for bad (yellow), less bad (blue) and good (red) layouts

In fig. 8 we see that the overall waveform of the scope trigger is largely the same, with minor differences in the peak-peak of the signals. There is a peak-peak voltage of 3.3v 3.9v and 5.1v for the bad, less bad and good layouts, respectively. Consequently, we see the lowest rise time (4.34 ns) for the bad layout and fastest (2.06 ns) on the good layout.

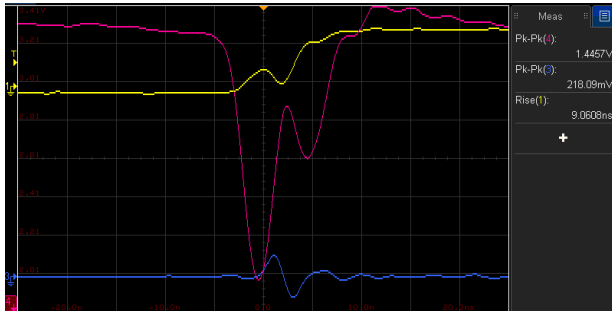
Moving forward to the HIGH and LOW readings on each layout, the scope trigger signal for that layout will be shown in channel one (yellow) at an offset to illustrate better the noise induced by the trigger signal.

Note: All of the following figures will have the HIGH in red and LOW in blue

7.3. Bad Layout: No Common Return Plane, Badly Placed Decoupling Capacitor



(a) Ground truth HIGH and LOW (500mV/div)



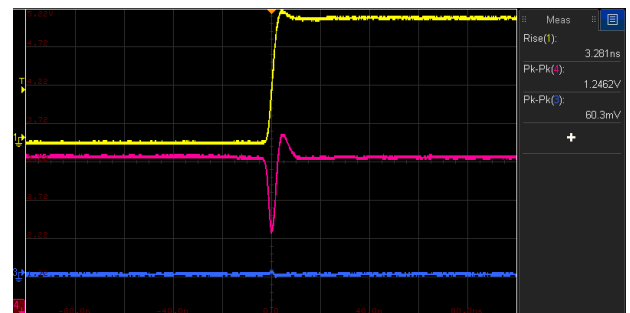
(b) Board HIGH and LOW (200mV/div)

Figure 9. Quiet HIGH and LOW readings on the bad layout from ground truth and assembled board

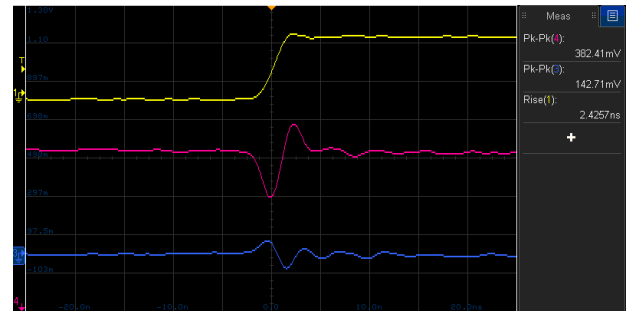
From fig. 9 we see that the assembled board replicates the HIGH reading from the ground truth, but does not replicate the LOW reading. Another thing to make note of is that the assembled board has a much slower rise time than the ground truth. This difference in rise time is the most likely reason why the LOW reading was not replicated on the assembled board reading.

Despite this, both boards demonstrate that the lack of a well-placed decoupling capacitor and common return plane induced a significant amount of noise on the p (positive) line and m (minus) line.

7.4. Less Bad Layout: Common Return Plane, Badly Placed Decoupling Capacitor



(a) Ground truth HIGH and LOW (500mV/div)



(b) Assembled board HIGH and LOW (200mV/div)

Figure 10. Quiet HIGH and LOW readings on the less bad layout from ground truth and assembled board

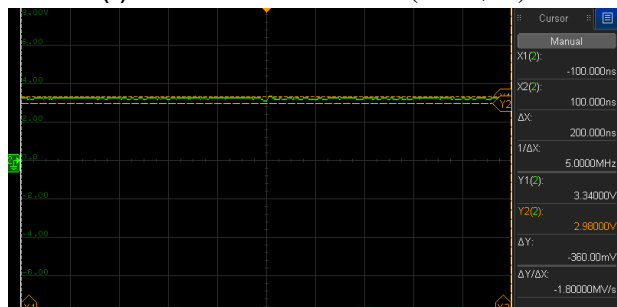
From fig. 10 we see that the assembled board has drastically different measurements, with the voltage drop being .5x-4x smaller than the drops measured in the ground truth. Additionally, the ground truth illustrates that the inclusion of the ground plane does better than the assembled board; including the ground plane drastically improved the amount of noise seen on the LOW signal (about a 7-8x reduction in noise) while the noise on the HIGH pin was reduced by nearly 5x.

However, the amount of noise on the ground and positive lines are similar to the assembled board. Comparing the bad layout and less bad layout is not as productive as well since the scope triggers are different.

7.5. Good Layout: Common Return Plane, Well Placed Decoupling Capacitor



(a) Ground truth HIGH and LOW (500mV/div)



(b) Assembled board HIGH



(c) Assembled board LOW

Figure 11. Quiet HIGH and LOW readings on the good layout from ground truth and assembled board (2V/div)

Note: The reason for the difference in presentation is that the measurements for the good layout on the assembled board were taken at different times. Figures 11b and 11c still show noise on the HIGH and LOW pins.

From fig. 11 we see that it is not entirely clear if the assembled board replicated the results from the ground truth. It would be more beneficial to zoom in on the signal on the switch of the scope trigger, like in previous figures. However, it is clear that both boards, including a return plane and a well-placed decoupling capacitor reduce the noise generated on the HIGH and LOW to a negligible amount.

8. Conclusion

The purpose of this lab was to familiarize ourselves with the impact of good and bad design choices. To do this, we designed a hex-inverter circuit driven by a 555-timer clock. Three iterations of this board were created, one with a badly placed decoupling capacitor and without a return plane (bad layout), one with a badly placed decoupling capacitor and with a return plane (less bad layout), and one with a well-placed decoupling capacitor and with a return plane (good layout). This way, the specific design practice and their effect on the board can be analyzed.

Going off of the criteria we set in the POR, the board assembled in this lab does work. The board receives power and the necessary

clock signals from the 555 and hex-inverters. The noise on the power lines is worse on the bad layout and negligible on the good layout, indicating that we were able to take advantage of our learnings about good design practices to create poor signal integrity by making a badly designed circuit.

However, the assembled board was not able to properly recreate the ground truth in some cases. The ground truth clearly demonstrates the different effects a return plane and decoupling capacitors have on the power lines, something that could not be said with the assembled board. Although the board was able to demonstrate poor signal integrity with poor design, the fact that it fails to show the specific effects of certain design choices is a flaw of the board. This is likely caused by poor soldering with the 555 timer circuit, since the scope trigger signal varied from layout to layout (more than what is expected).

The key takeaways are as follows:

1. Ensure that key circuit blocks are functioning before doing a full test of the board. This way, it will be clear which block is causing discrepancies between the expected measurements and the actual measurements.
2. Double-check the soldering of circuit blocks. It is possible that too much liquid flux is being used. Despite being no-clean flux, there is still residue left over between the solder mask and components which may act as insulation and interfere with measurements.
3. The inclusion of a common return plane drastically reduces the noise on the ground line of the board. Return planes should always be included in boards moving forward as it is a cheap and effective way to improve signal integrity.
4. The inclusion of decoupling capacitors drastically reduces the noise on the power line of the board. This way, the power given to other components isn't influenced by noise, thus guaranteeing component function and improved signal integrity.